AgriFlow Foods

Agricultural Supply Chain Analytics — Learner Capstone

1. Engagement Context

You are supporting a farm-to-customer supply-chain transformation for AgriFlow Foods. The business sources grains, fruits and vegetables from farms and cooperatives, transports them through ambient and cold-chain networks, holds lot-level inventory, and serves retailers, processors, HORECA, wholesalers and export customers.

The executive team wants to improve margin, service and food-safety resilience without simply carrying more inventory or paying more for supply.

2. End-to-End Physical Flow

Farm → Harvest Lot → Quality Inspection → Procurement → Inbound Transport → Warehouse / Cold Storage → Lot Allocation → Customer Order → Outbound Delivery → Customer.

3. Management Questions

- Which farms deliver the best total value after yield, quality and price are considered together?
- Where is cold-chain failure creating avoidable physical and financial loss?
- Which warehouses/products have the greatest spoilage and shelf-life risk?
- Why are fill rate and OTIF below target, and what should change in S&OP/inventory policy?
- How much safety stock is justified by demand variability and lead time?
- Can a suspect lot be traced from farm to every affected customer quickly?
- Which interventions produce the highest risk-adjusted financial benefit?

4. What Is New in This Project

- Lot-level farm-to-customer traceability and simulated recall containment.
- Perishability, shelf-life and FEFO-style dispatch risk analysis.
- Cold-chain temperature excursions, reefer compliance and transport loss.
- Expected vs actual agricultural yield and weather/irrigation risk.
- Procurement price variance against monthly market benchmarks.
- Demand forecasting with a true holdout period and WAPE.
- Safety stock and reorder-point calculations using demand variability and lead time.
- Working-capital vs spoilage trade-offs.
- Total-delivered-cost thinking: farm price alone is not enough.
- Scenario-based investment sizing with explicit low/base/high capture rates.

5. Dataset Scale

Dataset	Rows / Scale	Purpose
Farms	3,000	Supplier, irrigation, certification, primary crop
Harvest lots	22,000	Yield, price, quality, acceptance, expiry

Inbound shipments	~21,000	Carrier, delay, cold chain, freight, transport loss
Lot inventory	~21,000	Ageing, shelf life, spoilage and closing stock
Customer orders	30,000	Demand, fill rate, OTIF, rejection and sales
Lot allocations	30,000	Traceability from customer order back to harvest lot
Products	14	Ambient, cool and cold products across grains/fruit/vegetables

6. Required Workflow

Phase	Question	Output
Frame	What decisions must management make?	Issue tree and hypotheses
Validate	Can quantity, price, quality and lineage be trusted?	DQ issue log
Source	Which farms should receive more/less volume?	Supplier scorecard
Move	Which carriers/lanes damage cold-chain performance?	Carrier scorecard
Store	Where is ageing/spoilage concentrated?	Shelf-life & warehouse analysis
Serve	Where do service/margin failures occur?	Fill, OTIF, rejection, landed-cost analysis
Plan	How should demand/inventory be planned?	Forecast, WAPE, safety stock, reorder points
Respond	Can a contaminated lot be contained?	Recall impact list
Size & Recommend	What should management fund first?	Excel business case + executive memo

7. Deliverables

- PostgreSQL database loaded from raw files and cleaned reproducibly.
- 50 SQL business questions.
- Python notebook with supplier/logistics/inventory analysis, forecasting and traceability.
- Power BI 7-page supply-chain control tower.
- Excel opportunity-sizing model.
- Executive recommendation memo with a 90-day initiative plan.

8. Submission Standard

Every major insight should follow Evidence → Root Cause → Operational/Financial Impact → Recommendation. Do not treat an addressable loss pool as guaranteed savings. State assumptions and confidence explicitly.

All company names and records are synthetic. The workflow is inspired by common consulting and industry analytics practices; it does not reproduce proprietary methods of EY, Bain, McKinsey, or another firm.